14	Ammonium molybdate test: To one portion of the extract , add dil HNO_3 until the effervescence ceases, then add about 1mL each of ammonium molybdate and Conc. HNO_3	A canary yellow ppt is formed.	Presence of phosphate
15	Test with sodium nitro bruside: To about 1mL of the sodium carbonate extract add 1mL of dil .aommonia. Then add about few drops of sodium nitro prusside.	A purple or violet colouration appears	Presence of sulphide.

Reasoning

1	Action of heat: The reddish brown gas is NO_2 and N_2O_4 The zinc sulphate salts are yellow when hot and white when cold due to formation of semiconducting ZnO. At high temp there is bandgap excitation of electrons. When the excited electrons fall back from valence band to conduction band, light is emitted.
2	Flame test: To convert metallic salts into chlorides Conc.HCl is used. Generally chloride salts are more volatile than sulphate salts, so sulphuric acid is not used. When the metallic chlorides are introduced into flame, they are vapourised. In the middle of the flame, Cl- reduces metal ions into metal atoms. The metal atoms are then excited by the flame photons, and when the excited electrons fall back, they emit light of characteristic colour.
3	Copper turning test: When copper turning dissolves in sulphuric acid, electrons and copper ions are formed. The electrons are then used to reduce nitrate in the presence of H^+ to NO2
4	Chromyl chloride test: When a chloride salt is heated in the presence of sulphuric acid, CrO_2Cl_2 chromyl chloride is formed. With NaOH it forms Na_2CrO_4 .With lead acetate it forms yellow $PbCrO_4$ ppt
5	Test for halides: If any halide ion is present in sodium carbonate extract, it reacts with silver nitrate to form , to form the silver halide ppt. chloride forms AgCl ppt, bromide forms AgBr ppt,and iodide forms AgI ppt. AgCl dissolves in ammonia by forming $Ag(NH_3)_2^+$ complex. AgBr forms little amount of complex

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